

Console-Based Java Expense Tracker with Login System

Mini Project Report

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Year: 2025

1. Abstract

This project is a console-based Expense Tracker developed using Java. It allows users to log in securely and manage personal expenses by adding titles, amounts, and dates. All expense data is stored in a local file, enabling persistent storage even after the program exits. This project demonstrates Java fundamentals like file handling, object-oriented programming, and user interaction via the console.

2. Objective

The main objective of this project is to build a simple, secure, and user-friendly console-based application to track daily expenses and understand how Java can be used for real-world applications.

3. Technologies Used

- Programming Language: Java
- Environment: GitHub Codespaces / IntelliJ IDEA
- File Handling: Text File (expenses.txt)

4. System Design

The system is composed of four main classes:

- Main.java: Entry point and menu interface
- UserManager.java: Handles login validation
- Expense.java: Model class representing an expense
- ExpenseManager.java: Business logic for adding, viewing, and totaling expenses

5. Features

- User login with fixed credentials
- Add expenses with title, amount, and date
- View all added expenses
- Calculate total expenses
- Persistent storage using file system

6. Code Modules Explained

1. Main.java – Provides menu options and handles user input.
2. UserManager.java – Validates login using hardcoded username/password.
3. Expense.java – Represents an individual expense as an object.
4. ExpenseManager.java – Reads and writes data to 'expenses.txt', handles logic.

7. Sample Input/Output

Input:

Username: admin

Password: 1234

Choice: 1

Title: Groceries

Amount: 500

Date: 2025-07-24

Output:

Expense added.

Choice: 2

Title: Groceries, Amount: ₹500, Date: 2025-07-24

8. Conclusion

This Java mini project provided practical experience in developing a full-featured console application. It reinforced the importance of clean code structure, file operations, and the basics of secure user input handling.

9. Future Enhancements

- Add support for multiple user accounts
- Add graphical user interface using JavaFX or Swing
- Export reports to PDF or Excel
- Monthly/Category-based analytics
- Password hashing and file encryption